

Cycle-Slip Resilient High-Precision Transmitter IQ Skew Estimation for Ultra-High Baud-Rate Digital Subcarrier Multiplexing Transmission

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Abstract We propose a cycle-slip resilient high-precision transmitter IQ skew estimation scheme based on a designated training sequence. The 120-GBaud DP-16QAM digital subcarrier multiplexing (DSM) experimental transmission verifies its effective suppression of phase-extraction cycle slips, improving IQ skew estimation accuracy from 0.3 ps to below 0.1 ps. ©2026 The Author(s)

Introduction

The rapid growth of applications such as cloud computing, artificial intelligence, and 6G has driven optical networks toward higher baud-rate and larger capacity^{[1],[2]}. In coherent fiber optical transmission systems, advanced modulation formats combined with digital signal processing (DSP) are essential to fully utilize bandwidth and enhance transmission performance. Due to its high spectral efficiency^[3] and robustness against fiber nonlinearity^[4] and equalization-enhanced phase noise^[5], digital subcarrier multiplexing (DSM) outperforms single-carrier counterpart. However, DSM is more sensitive to transceiver imperfections such as IQ skew^{[6],[7]}. With the growing baud rate and modulation format order, IQ skew significantly degrades DSM transmission performance^[8], making a high-precision estimation of IQ skew critical.

Traditional IQ skew estimation methods rely on multiple-input-multiple-output (MIMO) equalizers. For instance, a single-tap 4×4 real-value (RV) MIMO adaptive equalizer is employed to process the I/Q tributaries of symmetric subcarrier pairs, utilizing the converged equalizer coefficients to estimate the IQ skew^[9]. A multi-tap 4×4 RV MIMO equalizer for joint equalization of symmetric subcarriers is also used to estimate the IQ skew, by extracting the phase values of the left and right subcarriers, exploiting their odd-symmetric property^[10]. However, such MIMO methods typically suffer from high computational complexity, significant hardware resource overhead, and limited estimation accuracy. Furthermore, IQ skew can be estimated by using a designated training sequence (TS). A symmetric subcarrier TS, where the first half is negative conjugate and the second half is positive conjugate, is designed to estimate IQ skew by extracting frequency-domain (FD) phase values^[11]. However, the estimation accuracy of this method is susceptible to frequency

offset (FO) and phase noise (PN). To address this issue, the same TS can be imposed on paired subcarriers to suppress interference from FO and PN, thereby improving IQ skew estimation performance^[12]. Nevertheless, those TS approaches assume favourable transmission conditions and suffer from severe performance penalty especially, under the condition of low optical signal-to-noise ratio (OSNR), where noise may induce phase-unwrapping cycle slips, leading to estimation failure during the phase-extraction.

In this submission, we present a cycle-slip resistant, high-precision transmitter (Tx) IQ skew estimation scheme for ultra-high baud-rate coherent DSM systems, based on the designated TS. Particularly, the method employs a Hampel filter (HF) to remove the noise-induced phase outliers in the received TS, preventing phase-extraction cycle slips and enhancing IQ skew estimation. Experimental verification of 120 GBaud DP-16QAM DSM signals with 4 subcarriers (SC) transmission over 100 km standard single-mode fiber (SSMF) demonstrate both suppression of cycle slips and accuracy improvement of IQ skew estimation from 0.3 ps to below 0.1 ps.

Principle

As shown in Fig. 1(a), we insert a specially designed TS before the payload. The first half of this TS satisfies a negative conjugate relationship between symmetric subcarriers, while the second half satisfies a positive conjugate relationship. Non-symmetric paired subcarriers, on the other hand, employ same sequences. Fig. 1(b) illustrates the overall process of the proposed scheme for estimating IQ skew after the DSM signal demultiplexing at the receiver. Each subcarrier signal, after demultiplexing and down-conversion to baseband, can be denoted as $r_p^\varepsilon(n)$, where p represents the subcarrier index and ε denotes

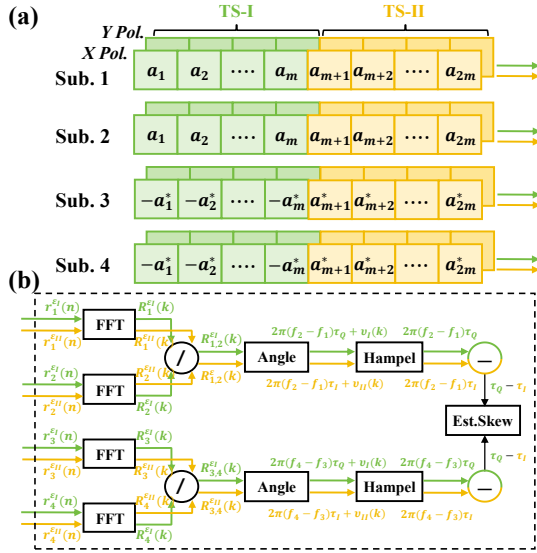


Fig. 1: (a) Design of the training sequence; (b) the proposed IQ skew estimation scheme.

the X or Y polarization state. For the ease of illustration, taking the X polarization as an example, performing a fast Fourier transform (FFT) on $r_p^x(n)$ yields the FD representation as^[12]

$$R_p^x(k) = \frac{1}{2} \left\{ [\alpha_I e^{-j(\omega+2\pi f_p)\tau_I} + \alpha_Q e^{j\varphi} e^{-j(\omega+2\pi f_p)\tau_Q}] X_p(\omega) + [\alpha_I e^{-j(\omega+2\pi f_p)\tau_I} - \alpha_Q e^{j\varphi} e^{-j(\omega+2\pi f_p)\tau_Q}] X_q^*(-\omega) \right\} \otimes \Lambda(\omega) \quad (1)$$

where τ_I , τ_Q , α_I , α_Q , φ denote the time delays of the I and Q tributaries, amplitude gains, and phase imbalance, respectively. f_p represents the central frequency of the p -th subcarrier, q is the index of the subcarrier symmetric to p , and k is the frequency index. The symbols '*' and ' $\otimes$ ' represent the conjugate operation and convolution operation, respectively. $X_p(\omega)$ and $X_q^*(-\omega)$ denote the FD representations of the initially indexed subcarriers p and q under the X polarization, which are symmetric to each other. $\Lambda(\omega)$ is the FD representation of the phase noise.

Since the first half of the TS satisfies the negative conjugate relationship between symmetric subcarriers, i.e., $x_p(n) = -x_q^*(n)$, it follows that $X_p(\omega) = -X_q^*(-\omega)$. In such case, the received signal of each subcarrier in the FD can be further expressed as

$$R_p^{X_I}(k) = \alpha_Q e^{j\varphi} e^{-j(\omega+2\pi f_p)\tau_Q} \cdot X_p(\omega) \otimes \Lambda(\omega) \quad (2)$$

Similarly, the second half of the sequence satisfies the positive conjugate relationship, i.e., $X_p(\omega) = X_q^*(-\omega)$, and the FD representation of each subcarrier signal becomes

$$R_p^{X_{II}}(k) = \alpha_I e^{-j(\omega+2\pi f_p)\tau_I} \cdot X_p(\omega) \otimes \Lambda(\omega) \quad (3)$$

For non-symmetric paired subcarriers, their TS are same. Taking subcarriers 1 and 2 as an

example, where $X_1(\omega) = X_2(\omega)$, dividing one by the other yields

$$R_{1,2}^{X_I}(k) = \frac{R_1^{X_I}(k)}{R_2^{X_I}(k)} = e^{j2\pi(f_2-f_1)\tau_Q + v_I(k)} \quad (4)$$

$$R_{1,2}^{X_{II}}(k) = \frac{R_1^{X_{II}}(k)}{R_2^{X_{II}}(k)} = e^{j2\pi(f_2-f_1)\tau_I + v_{II}(k)} \quad (5)$$

where $v_I(k)$ and $v_{II}(k)$ represent outliers caused by noise. Under the condition of low OSNR, noise-induced outliers become more pronounced, generating multiple spikes in the extracted phase values and severely degrading IQ skew estimation accuracy. Furthermore, directly performing phase unwrapping under those conditions is highly prone to inducing cycle slips^[13]. To address this issue, HF is applied to the phase values^[14], in order to remove outliers, effectively preventing cycle slips and achieving high-precision IQ skew estimation. After the HF processing, the estimated IQ skew is expressed as

$$\Delta\tau_{skew} = \tau_Q - \tau_I = 1 / 2\pi(f_2 - f_1) \cdot \text{Hampel} \left\{ \begin{array}{l} \text{unwrap}[\text{angle}(R_{1,2}^{X_I}(k))] \\ -\text{unwrap}[\text{angle}(R_{1,2}^{X_{II}}(k))] \end{array} \right\} \quad (6)$$

where $\text{angle}(\cdot)$, $\text{unwrap}[\cdot]$, and $\text{Hampel}\{\cdot\}$ denote the operators for phase angle extraction, phase unwrapping, and HF, respectively. Similarly, the same processing can be applied to subcarriers 3 and 4 to estimate the IQ skew, and the overall flowchart is shown in Fig. 1(b). This approach combines TS-based structure with HF processing, providing robust, cycle-slip resistant Tx IQ skew estimation even under poor OSNR conditions, and significantly improving estimation accuracy to the sub-ps level.

Experimental Results

The experimental setup, in together with Tx/Rx DSP flow, is shown in Fig. 2(a). A tunable laser operated at 1550.12 nm is used to generate optical carrier, with an output power of 16 dBm and a linewidth of 100 kHz. The electrical signal is generated by an arbitrary waveform generator (AWG) with a 3 dB bandwidth of 70 GHz and sampling rate of 256 GSa/s, whose outputs are directly introduced to a coherent driver modulator (CDM). The modulated optical signal is then amplified by an erbium-doped fiber amplifier (EDFA) and transmitted over 100 km SSMF. The OSNR is adjusted by jointly tuning a variable optical attenuator (VOA) and an EDFA. At the receiver, another tunable laser acts as local oscillator (LO). After coherent detection, the electrical signals are captured by a real-time oscilloscope (RTO) operated at 256 GSa/s for subsequent offline processing. Fig. 2(b) and (c) depict the spectrum of the transmitted and received 4 subcarriers 120 GBaud

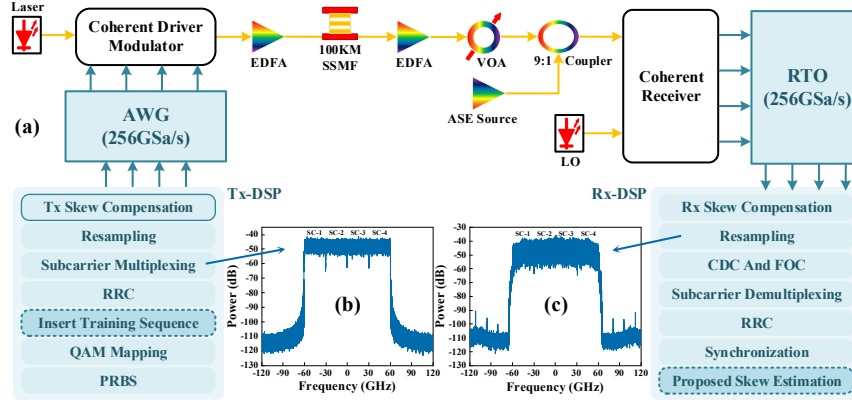


Fig. 2: (a) Experimental setup and the corresponding Tx/Rx DSP flow; spectrum of the 120 Gbaud DP-16QAM 4SC-DSM signals are presented in (b) transmitted, (c) received.

DSM signals, respectively. For the Tx-DSP, a specifically designed TS are inserted before the payload of each subcarrier, for the purpose of high-precision estimation of IQ skew. As for the Rx-DSP, Rx IQ skew compensation is done first. And the consequent DSP flows include resampling, chromatic dispersion compensation (CDC), frequency offset compensation (FOC), subcarrier demultiplexing, root raised cosine (RRC) filter, synchronization, and the proposed IQ skew estimation.

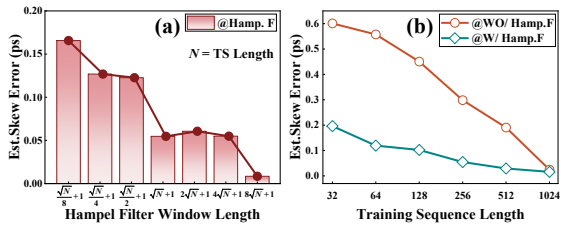


Fig. 3: IQ Skew estimation error versus (a) HF window length and (b) TS length.

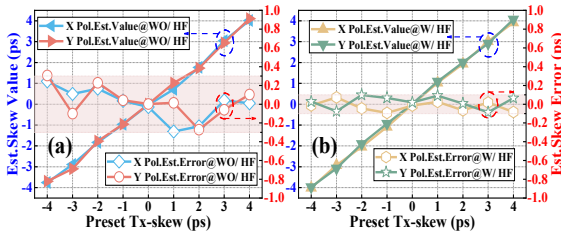


Fig. 4: IQ skew estimation performance versus schemes for (a) without HF and (b) proposed with HF.

Fig. 3(a) shows the impact of the HF window length L , which adapts with the TS length N , on the IQ skew estimation accuracy under conditions of an OSNR of 25 dB and a preset IQ skew of 1 ps. The results indicate an optimal setting of $L = \sqrt{N} + 1$. Fig. 3(b) compares IQ skew estimation accuracy versus the TS length. Without the use of HF, estimation errors remain high, particularly for short TS, due to noise-induced outliers. When HF is applied, errors are consistently reduced, and the estimation stabilizes when the TS length reaches 256. To achieve robust, high-precision IQ skew estimation, we set the TS length to 256 for all subsequent experiments.

Fig. 4(a) and (b) demonstrate the estimation

performance for IQ skew without and with HF, respectively, at an OSNR of 27 dB as the preset skew varies within ± 0.5 symbol period. Experimental results indicate that, the conventional scheme without HF, affected by noise-induced outliers, achieves an estimation accuracy within ± 0.3 ps. In contrast, the proposed scheme with HF improves accuracy to within ± 0.1 ps, when the skew varies from -4 to 4 ps.

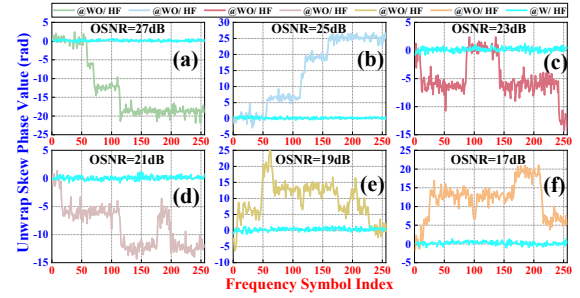


Fig. 5: Cycle slips with versus without HF for OSNR (a) 27 dB, (b) 25 dB, (c) 23 dB, (d) 21 dB, (e) 19 dB, (f) 17 dB.

Figs. 5 illustrates the occurrence of cycle slips in the estimated IQ skew phase after the phase unwrapping, when the OSNR range is from 17 dB to 27 dB. Without the HF, pronounced cycle slips are observed, due to noise-induced outliers. After applying HF, those outliers are effectively removed, eliminating cycle slips and demonstrating the cycle-slip resilient performance of the proposed scheme over the OSNR range.

Conclusions

We have proposed and experimentally verified a cycle-slip resistant, high-precision Tx IQ skew estimation scheme, when the 120 Gbaud DP-16QAM DSM signals with 4 subcarriers are transmitted over 100 km SSMF. By removing noise-induced outliers with a HF, our proposed scheme eliminates phase cycle slips over an OSNR range from 17 dB to 27 dB, and significantly improves estimation robustness, enhancing the IQ skew estimation accuracy from 0.3 ps to below 0.1 ps. Those results validate precise IQ skew compensation for ultra-high baud-rate DSM transmission systems, advancing next-generation high-speed optical networks.

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